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ORIGINAL ARTICLE

Efficacy of various insecticides recommended for indoor residual spraying: pirimiphos methyl, potential alternative to bendiocarb for pyrethroid resistance management in Benin, West Africa

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Background: Using the same insecticide for multiple successive indoor residual spraying (IRS) cycles is not recommended; instead, the National Malaria Control Program (NMCP) has decided to select another insecticide (insecticide B) in addition to bendiocarb for indoor residual spraying.

Methods: An experimental hut trial comparing the effectiveness of three classes of insecticides (one carbamate [bendiocarb], two organophosphates [fenitrothion and pirimiphos methyl] and one pyrethroid [lambdacyalothrin]) was conducted in Malanville, northern Benin, against wild free entered resistant *Anopheles gambiae* s.l. population to pyrethroids.

Results: Fenitrothion and pirimiphos methyl yielded the highest rate of deterrence. Their mean rates were respectively 46.6% and 35.4%. Regarding blood feeding inhibition, only fenitrothion has induced a significant inhibition rate (25.4% as mean rate). As regards the exophily rates, only lambdacyhalothrin has induced the highest rate (39.7%). Pirimiphos methyl showed the highest mortality rate and also induced a mortality rate of at least 80% in blood fed *An. gambiae* population after 24 h observation time. Furthermore, the huts treated with pirimiphos methyl showed the highest residual effect, followed by lambdacyhalothrin.

Conclusion: Benin NMCP has selected pirimiphos methyl as insecticide B to alternate or combine to bendiocarb (carbamate) because of the adverse effects of fenitrothion on the sleepers and its short residual effect on walls.

Keywords: *Anopheles gambiae*, Benin, Experimental huts, Pyrethroid resistance

Background

In West Africa, *Anopheles gambiae*, *Anopheles coluzzii*[1] and *Anopheles arabiensis* are members of *An. gambiae* complex and constitute the main vectors of malaria. *An. gambiae* and *An. coluzzii* co-exist in West Africa[2] and transmit the Plasmodium causative agent of malaria thanks to its high anthropophily.[3]

Due to the resistance of Plasmodium to anti-malaria drugs and the lack of vaccines, one of the most effective controls of malaria is to limit the transmission. This includes the use of long lasting insecticide-treated bed nets (LLINs) and indoor residual spraying (IRS) of insecticides. Control of malaria vectors relies mainly on the use of insecticides. Among these, pyrethroids are the only compounds approved to impregnate bed nets and other netting material for personal protection. However, the expansion of pyrethroid resistance in the main malaria vectors constitutes a great threat to the implementation of LLINs and IRS in Africa.[4–6] A study conducted in experimental huts showed a reduced efficacy of lambdacyhalothrin treated nets against *An. gambiae* in Ladji, a peripheral area of Cotonou.[7]

IRS consists of the application of residual insecticides to the inner surfaces of dwellings where many vector species of *Anopheline* mosquito tend to rest after taking a blood meal.[7] It is effective in rapidly controlling malaria transmission thus reducing the local burden of malaria morbidity and mortality, provided that most houses and animal shelters (>80%) in targeted communities are treated.[8]

In Benin, *An. Gambiae*, has developed a high level of resistance to pyrethroid insecticides,[5] which is a serious concern to the future use of LLINs and IRS. In this context, it is urgent to investigate alternative classes of insecticides with different modes of action than pyrethroids. Hence, several studies showed the spread of insecticide resistance with high frequencies of *L1014F* to DDT and pyrethroids mainly across the southern areas.[9] The data

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from Corbel *et al*. suggest that the *1014S* mutation has emerged in *An. gambiae s.l.* in Malanville.[10] Two P450 genes *CYP6M2* and *CYP6P3,* have been found to be significantly over-expressed in permethrin resistant *An. coluzzii* from the south-east of Benin (Akron) following microarray analysis.[11]

In 2008, the National Malaria Control Program (NMCP) implemented a vector control intervention based on IRS of bendiocarb. Four districts of southern Benin with high resistance of *An. gambiae* to pyrethroids were sprayed with bendiocarb; more than 350 000 inhabitants have been protected.[12] Entomological parameters in the control area were compared with those in intervention sites. The study has shown a drastic decrease in *An. gambiae* biting rate and entomological inoculation rate in the sprayed areas. Bendiocarb was found to be a good alternative insecticide for IRS in Benin.[12]

According to the resistance monitoring policy in Benin, the NMCP has decided to select another insecticide (insecticide B) after the IRS intervention currently undergoing in Atacora with the support of the US President's Malaria Initiative (PMI). While waiting for the release of new insecticide molecules, the current priority for malaria programs in Africa is to maximize the effectiveness of existing insecticides. This is why Benin NMCP proposed to test various insecticides recommended by WHO for IRS. Using the same insecticide for multiple successive IRS cycles is not recommended; instead, it is preferable to use a system of rotation with a different insecticide class being used each year. There are several approaches to management of insecticide resistance, but all rest on the minimization of the pressure of selection for the development of resistance, in limiting the prolonged exposure of populations of mosquito to insecticides. The goal of this study is to evaluate and select, from among recommended insecticides, another effective insecticide B in addition to bendiocarb for carbamate resistance management.

Methods

Study site and experimental huts

The study was implemented in Malanville, northern Benin. Malanville district is characterized by the presence of a 100 hectares rice growing area which is located at 11°52′ N and 3°23′ E near the Niger River. Malanville is characterized by a Soudanian climate (semiarid) with only one rainy season per year; the main annual rainfall is 900 mm. *An. gambiae,* the main malaria vector in Malanville, is resistant to pyrethroids with a kdr frequency more than 0.6. *An. gambiae coluzzii* is predominant in Malanville. Five experimental huts were selected for the study. These huts are typical of the West African region and are made from concrete bricks, with a corrugated iron roof, a ceiling of thick polyethylene sheeting, and a concrete base surrounded by a water-filled channel to prevent entry of ants (Figure 1). Mosquito access is via four window slits constructed from pieces of metal fixed at an angle to create a funnel with a 1 cm wide gap. Mosquitoes fly upward to enter through the gap and downwards to exit; this precludes or greatly limits exodus through the aperture enabling the majority of entering mosquitoes to be accounted for. A single veranda trap made of polyethylene sheeting and screening mesh measuring 3 m long, 2.5 m wide, and 1.5 m high, projects from the back wall of each hut. Movement of mosquitoes within huts and veranda is unimpeded during the night.



Figure 1. Experimental hut station of Malanville. These huts are typical of the West African region and are made from concrete bricks, with a corrugated iron roof, a ceiling of thick polyethylene sheeting, and a concrete base surrounded by a water-filled channel to prevent entry of ants

Insecticide treatment

The insecticides used per labelled hut at the WHO recommended dose were: Hut 1 (15.86 m^2): Ficam VC [Bayer (Pty) Ltd, Isando, South Africa] at 80% (bendiocarb: carbamate; 0.4 g/m^2); Hut 2 (17.21 m^2): SUMITHION 40WP (Sumitomo, Perai, P. Pinang, Malaysia) (fenitrothion organophosphate; 2 g/m^2); Hut 3 (17.19 m^2): unsprayed, control; Hut 4 (17.33 m^2): Icon 10 CS (Syngenta Crop Protection AG, Basel, Switzerland) (100 g/L), lambdacyalothrin: pyrethroid; (0.03 g/m^2); Hut 5 (17.16 m^2): Actellic 300 CS (Syngenta experimental compound, Basel, Switzerland) (1.06 g/cm^3), pirimiphos methyl: organophosphate (2 g/m^2). The absorbency of the walls was 124 mL/m^2. Indoor residual treatments were applied with a hand-operated compression sprayer. Only the interior cement walls were sprayed uniformly after masking the veranda and window slits with protective coverings. The evaluation started 48 hours after treatment and ran for 6 months from 29 September 2011 to 29 March 2012.

Sleepers and mosquito collections

An announcement was made in the district that the study was looking for volunteers. They were recruited among the inhabitants of the location in Malanville (pregnant and breast feeding women were not included in the study). With the support of the traditional head of district, the sleepers were chosen via a selection process and were selected with one supervisor. They were informed on the objective of the study and signed an informed consent for those who were literate or gave their oral consent for those who were illiterate.

A preliminary experiment showed the huts to be equally attractive to mosquitoes. The treatments were randomly allocated to the five experimental huts. Five adult men slept overnight

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in the huts from 21:00 to 06:30 h and collected mosquitoes from the huts in the morning.

The sleepers were rotated between huts to correct for possible variation in individual attractiveness. Each morning, mosquitoes were collected from the floor, walls, ceilings and verandas using tubes, cotton and torches. Mosquitoes were identified and scored as blood fed or unfed and dead or alive. Live mosquitoes were held in netted plastic cups and supplied with 10% honey solution for 24 h before delayed mortality was recorded. Male mosquitoes were not recorded. Some live mosquitoes were analyzed using standard PCR for kdr mutation detection.

The entomological impact of each treatment on mosquitoes is expressed relative to the control in terms of the following: Deterrence (reduction in hut entry relative to the control huts); Induced exophily (the proportion of mosquitoes that exited early and found in exit traps); and blood feeding inhibition (the reduction in blood feeding compared with that in the control huts). The number of blood fed mosquitoes (personal protection) relative to the control hut was calculated as follows: % personal protection=$100(B_u-B_t)/B_u$ where B_u is the number of blood fed mosquitoes in the untreated control hut and B_t is the number of blood fed mosquitoes in the huts with insecticide treatments; Immediate and delayed mortality (the proportion of mosquitoes that are killed).

Residual activity of insecticide treatment

WHO cone bioassay was undertaken in each hut to evaluate the residual activity of the insecticides. At least 10 to 15 females of a susceptible laboratory strain of *An. gambiae* (Kisumu) aged 3–5 days were introduced into four cones attached to each face of walls per hut for 30 min exposure according to WHO guidelines.[7] Honey solution was provided during the 24 hour holding period at room temperature, 25°C.

Data analysis

Data were entered in Excel and transferred to R software[13] for further analysis. The numbers of mosquitoes collected each night were compared between treatments using a Wald's statistical test and also between the treatment huts and the untreated hut. Proportion data were analyzed using logistic regression after adjusting for the effects of sleeper attractiveness and hut position. The adjustment was done using Kruskal–Wallis rank sum test on the numbers of mosquitoes collected in each hut prior to treatment.

Results

Status of vector resistance at the beginning of the study

At the beginning of the study, *An. gambiae s.l.* larvae and pupae were collected from the study area and WHO susceptibility tests were performed with adult mosquitoes to measure the level of *An. gambiae* resistance to pyrethroids (deltamethrin and permethrin), bendiocarb, fenitrothion and to pirimiphos methyl. The high frequency of kdr mutation (F[kdr]) observed after PCR analyses on some live mosquitoes caught in experimental huts in Malanville (F[kdr]=0.9) confirmed the rapid spread of pyrethroid resistance in this district with the low mortality rate (23.7 and 3.9%) after exposure of females of *An. gambiae* respectively to deltamethrin and permethrin impregnated papers. According to WHO criteria,[14] the Anopheles mosquito population from Malanville was resistant to pyrethroids, but was susceptible to carbamates and organophosphates (Table 2).

Phase II evaluation in experiment huts

Over the six-month trial, 1751 free entry *An. gambiae s.l.* from Malanville, 3045 *Mansonia africana*, 300 *An. pharoensis*, *90 Cx. nebulosus*, five *An. ziemanni*, two *Culex quinquefasciatus*, three *An. nili* and two *An. funestus* were collected in the huts. *An. gambiae* and *An. funestus* are the main vectors implicated in malaria transmission. In Benin, the secondary known vectors of malaria (*An. pharoensis* and *An. ziemanni*) were never found implicated in the transmission of Plasmodium.

Attractiveness of huts before treatment

Before treating the huts with the various insecticides, mosquito collections were done inside to measure the individual attractiveness of each hut for mosquitoes. This study was carried out in September before starting the evaluation. The densities of mosquitoes observed in the five huts were compared. The homogeneity of attractiveness of the experimental huts was assessed using Kruskal–Wallis rank sum test on the numbers of mosquitoes collected in each hut prior to treatment. This analysis showed that the individual huts did not significantly differ in terms of the number of mosquitoes entering (p=0.8736 for *An. gambiae s.l.* and p=0.9066 for the culicine mosquitoes).

Entomological impacts of treatments

Anopheles gambiae

The summary results of the efficacy of each treated hut against host-seeking *An. gambiae s.l.* are presented in Table 1 and in

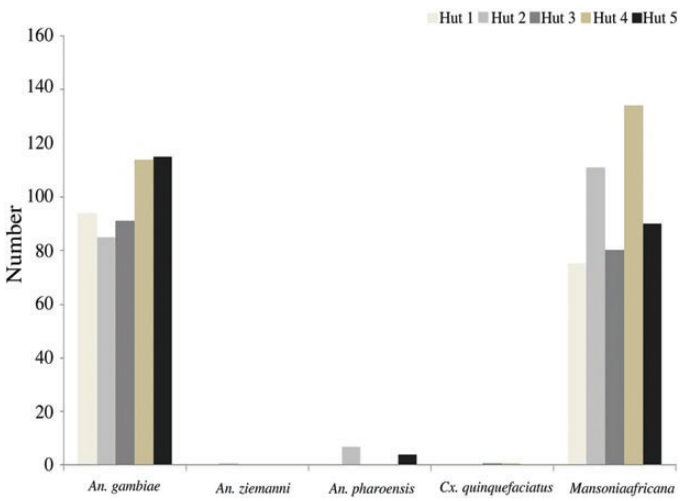


Figure 2. Number of *Anopheles.gambiae* and culicinae mosquitoes collected in five days in the experimental huts in Malanville before insecticidal treatment (sleepers spent the same number of nights in each hut). H: hut.

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Table 1. Summary of results obtained for *Anopheles gambiae* of Malanville in experimental huts (6 months data collected after treatment)

Treatments	Month	Total females caught	Deterrency (%)	Exophily (%)			Blood feeding (%)			Immediate corrected mortality(%)		Overall corrected mortality (%)		
				Rate	95% CL	Induced exophily (%)	Rate	95% CL	Blood feeding inhibition	Rate	p	Rate	95% CL	p
Untreated hut	1	152	-	17.11	11.12–23.09	-	95.39	92.06–98.73	-	0	-	0	0.00–2.46	-
	2	73	-	19.18	10.15–28.21	-	97.26	93.52–101	-	0	-	0	0.00–4.99	-
	3	79	-	17.72	9.30–26.14	-	97.47	94–100.93	-	0	-	0	0.00–4.64	-
	6	138	-	13.04	7.42–18.66	-	88.41	83.06–93.75	-	0	-	0	0.00–2.71	-
Bend	1	77	49.34	25.97	16.18–35.77	NS	94.81	89.85–99.76	NS	50.65	<0.001	97.4	93.85–100.00	<0.001
	2	88	−20.55	13.64	6.47–20.81	NS	87.5	80.59–94.41	10.04	5.68	0.04	7.95	2.30–13.61	0.01
	3	95	−20.25	29.47	7.61–21.86	NS	67.37	80.69–94.05	10.36	5.26	0.04	51.58	41.53–61.63	<0.001
	6	69	50	17.39	8.45–26.33	NS	92.75	86.64–98.87	NS	1.45	0.33	4.35	1.49–12.02	0.01
Feni	1	46	69.74	21.74	9.82–33.66	NS	80.43	68.97–91.90	15.68	100	<0.001	100	92.29–100.00	<0.001
	2	53	27.4	28.3	16.17–40.43	NS	67.92	55.36–80.49	30.16	69.81	<0.001	81.13	70.60–91.67	<0.001
	3	56	29.11	26.79	15.19–38.38	NS	67.86	55.63–80.09	30.38	66.07	<0.001	76.79	65.73–87.84	<0.001
	6	55	60.14	30.91	18.70–43.12	20.55	94.55	88.54–100.55	NS	1.82	0.28	9.09	1.49–16.69	<0.001
Lambda	1	176	−15.79	50	42.61–57.39	39.68	84.09	78.69–89.49	11.85	21.09	<0.001	92.05	88.05–96.04	<0.001
	2	118	−61.64	30.51	22.20–38.82	NS	93.22	88.68–97.76	NS	7.63	0.02	9.32	4.08–14.57	0.01
	3	124	−56.96	29.03	21.04–37.02	NS	93.55	89.22–97.87	NS	7.26	0.01	8.87	3.87–13.88	0.01
	6	85	38.41	20	11.50–28.50	NS	91.76	85.92–97.61	NS	3.53	0.03	3.53	1.21–9.87	0.03
Pirimiphos	1	75	50.66	17.33	8.77–25.90	NS	90.67	84.08–97.25	NS	96	<0.001	100	95.13–100.00	<0.001
	2	58	20.55	17.24	7.52–26.96	NS	89.66	81.82–97.49	NS	77.59	<0.001	91.38	84.16–98.60	<0.001
	3	60	24.05	16.67	7.24–26.10	NS	90	82.41–97.59	NS	76.6	<0.001	90	82.41–97.59	<0.001
	6	74	46.38	35.14	24.26–46.01	25.41	91.89	85.67–98.11	NS	28.38	<0.001	52.70	41.33–64.08	<0.001

Bend: bendiocarb; Feni: fenitrothion, Lambd: lambdacyhalothrin, NS: not significant; Pirimiphos: pirimiphos methyl; -: reference rate.
$p<0.05$: probability of significance; $p<0.001$ mean that the values are more significant than $p<0.05$.

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Table 2. Status of vector resistance at the beginning of the study in the experiment huts station of Malanville

Susceptibility test					Kdr mutation			
Tested insecticide	Locality	Tested number	Mean mortality rate	95% CI	RR	RS	SS	F (kdr)
Bendiocarb 0.1%	Malanville	93	100	96.11–100	38	10	0	0.90
Fenitrothion 1.0%		82	98.53	93.41–99.78	38	10	0	0.90
Pirimiphos methyl 0.25%		86	100	95.72–100	38	10	0	0.90
Permethrin 0.75%		97	3.93	21.93–41.12	38	10	0	0.90
Deltamethrin 0.05%		97	23.71	15.66–33.43	38	10	0	0.90

F (kdr): frequency of kdr mutation: RR: homozygote kdr resistant alleles; RS: heterozygote kdr alleles; SS: homozygote kdr susceptible alleles

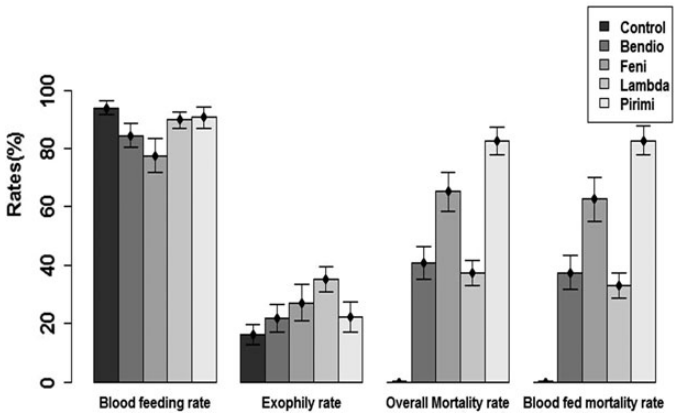


Figure 3. Entomological parameters measured with free *Anopheles gambiae* entry during the experiment in Malanville (WHOPES, phase II). The monthly data were cumulated parameter per parameter in treated and untreated huts. Blood fed mortality rate was calculated among blood fed population. Bendio: bendiocarb; Feni: fenitrothion; Lambda: lambdacyhalothrin; Pirimi: pirimiphos methyl.

Figure 3. There is some difference in the number of mosquitoes collected between huts (p=0.049). This difference cannot be due to the position of each hut because the collection done before the treatment of the houses (white catch of mosquitoes) has shown the same attractiveness of the huts for mosquitoes.

Significant deterrence (reduction of entry) of *Anopheles gambiae* in the huts treated with fenitrothion and pirimiphos methyl

The two huts treated with fenitrothion and pirimiphos methyl have shown a significant deterrence for *An. gambiae* compared to control ($p<0.05$). The deterrence rates fluctuated during the period as shown in Table 1 and the mean rate was high in the hut treated with fenitrothion (46.6%) compared to the hut treated with pirimiphos methyl (35.4%). However, the deterrence rate in the treated hut with bendiocarb was significantly higher compared to the control during the first month of the evaluation (49.3%; $p<0.05$; (Table 1).

Induced exophily observed from treatment

Only lambdacyhalothrin induced significant exophily rates on host-seeking *An. gambiae* in the first month: 50.0% versus 17.1% for the control, representing 39.7% of induced exophily ($p<0.05$; Table 1). Furthermore, fenitrothion and pirimiphos methyl have also induced exophily when we compared the mean rates ($p<0.05$).

High blood feeding inhibition with fenitrothion

The blood feeding inhibition by pirimiphos methyl and lambdacyhalothrin did not vary significantly from the control (respectively, OR 2.79:2.56 and 4.43 with $p>0.05$). Only fenitrothion induced a significant blood feeding inhibition: $p<0.05$. In the hut treated with pirimiphos methyl, blood feeding inhibition was not significant during the whole period with regard to the free entry of *An. gambiae* compared to the untreated hut (Table 1).

High immediate and overall mortality in huts treated with pirimiphos methyl and fenitrothion

During the first month, the immediate mortality rates were above 50% and the overall mortality rates were greater than 80% in all treated huts except the hut treated with lambdacyhalothrin. These mortality rates were significantly higher than that registered in the control hut ($p<0.05$). During the second month (Table 1), the overall mortality rates of wild *An. gambiae* in huts treated with fenitrothion and pirimiphos methyl were significantly higher (respectively 81.1% and 91.4% for wild *An. gambiae*). However these mortality rates progressively decreased after the first month until the sixth month where only pirimiphos methyl showed an overall mortality rate greater than 50%. As a whole, pirimiphos methyl displayed a significantly higher mortality rate during the experimentation (OR <1 and $p<0.001$). Moreover, in the huts treated with pirimiphos methyl, at least 80% of blood fed *An. gambiae* died after 24 hours observation holding time (differed mortality).

Insecticide residual activity

The residual activity of each insecticide in treated huts measured by cone bioassay tests with the laboratory strain Kisumu is

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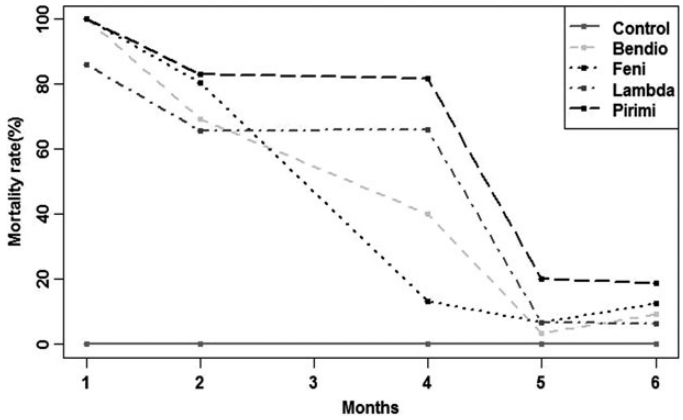


Figure 4. WHO cone bioassay: residual effect plot designed from different mortality rates recorded in treated huts per month. Laboratory susceptible *Anopheles gambiae* 'KISUMU' strain was used. Bendio: bendiocarb; Feni: fenitrothion; Lambda: lambdacyhalothrin; Pirimi: pirimiphos methyl.

illustrated by the mortality rate. This mortality rate was 100% for all treatments in the first month. The hut treated with pirimiphos methyl showed the better residual effect (4–5 months), followed by lambdacyhalothrin (3–4 months), bendiocarb (2–3 months) and fenitrothion (1–2 months; Figure 4).

Side effects of the treatment on sleepers

A regular follow up of side effects of the insecticide treatments on sleepers was conducted using a questionnaire. Certain complaints were registered during the evaluation when the sleepers spent the night in the hut treated with fenitrothion. Eye and nose irritation and treatment odours were reported. The main complaint mentioned by sleepers was fenitrothion odour which sometimes kept them awake. However, the sleepers noticed that the treatments reduced the biting nuisance of mosquitoes in the treated huts than in their own homes or the control hut. In response to the question 'Would you like to continue the experiment?' they all responded 'Yes'.

Discussion

Given the ever growing threat of insecticide vector resistance, the best strategy for controlling disease vectors remains the rotational use of insecticides of different modes of action altogether, rather than merely alternating members of the same chemical class or different chemical classes that address the same target site.[15] For example, the presence of kdr resistance renders DDT and pyrethroids less effective whereas carbamates, such as bendiocarb, or organophosphates can still be used in the absence of *Ace.1* mutation thus can be used in rotation. Such a strategy might increase the chance of regaining pyrethroid susceptibility, but it has to be carefully monitored so as to not select quickly resistant populations of vectors to insecticide.[16] In Benin, the NMCP has therefore decided to implement IRS in Atacora using the method of rotation or combination of insecticides that have different modes of action.

The reduction of entry of *An. gambiae s.l.* was the most evident factor observed in the huts treated with the organophosphates (fenitrothion and pirimiphos methyl) but lesser with carbamate (bendiocarb). Only pyrethroids are described as having repellent effects. But in this study, lambdacyhalothrin did not induce reduction of entry of *An. gambiae s.l.* during the whole experiment. This may be due to the treatment effects because the homogeneity of attractiveness of the experimental huts was assessed using Kruskal–Wallis rank sum test on the numbers of mosquitoes collected in each hut prior to treatment; this analysis showed that there was no significant difference in the number of mosquitoes entering the individual huts. The absence of the deterrent effect of lambdacyhalothrin against *An. gambiae s.l.* might be related to the high resistance of this species to pyrethroids[5,9,10] inducing the change in behaviour towards the repellent effect of pyrethroids.[17] Besides, a timed series of collections showed that the *L1014F* kdr allele increased in frequency from 6 to 80% in 6 years and as a result there was a drastic reduction in susceptibility to DDT and pyrethroids in the north part of Benin (Malanville). Likewise, in a previous study, the deterrent effect of repellents was not observed in *Culex quinquefasciatus*, a species known also to be much less responsive to the deterrent effect of pyrethroids.[18]

Unlike the absence of a deterrent effect of lambdacyhalothrin in *An. gambiae s.l.*, the induced exophily of wild *An. gambiae s.l.* was the highest in the huts treated with this insecticide resulting from the repellent effect of pyrethroids. However, the complete absence of efficacy of lambdacyhalothrin in *Culex quinquefasciatus* in Ladji merely confirms earlier findings involving other types of pyrethroid in experimental hut in West Africa.[19,20]

Moreover, lambdacyhalothrin has induced a significant blood feeding inhibition in the three first months in *An. gambiae s.l.*, likely because of its repellent effect. Regarding the other insecticides, fenitrothion has induced a significant blood feeding inhibition on wild *An. gambiae s.l.* This may be due to its high odour during the evaluation. The sleepers also complained about this side effect of fenitrothion. Nevertheless, low blood feeding inhibition was also observed in the huts treated with bendiocarb. But many phase II and phase III evaluations implemented in Benin continuously demonstrated that despite the treatment of the houses with insecticides, the majority of mosquitoes will successfully enter these treated houses and take their blood meal on their host before resting on the treated walls.[9,21] This finding explains why we have proposed to Benin NMCP to always invite communities who are protected by IRS to additionally sleep under LLINs to supplement and maximize malaria control efforts. However, such combination strategy implicates an increase in the cost of malaria prevention, but is desirable for areas with highest level of malaria transmission.[9]

During the first month, the immediate mortality rates were above 50% and the overall mortality rates were greater than 80% in all insecticides treatments except the huts treated with lambdacyhalothrin, and then were significantly higher than the rates recorded in the control ($p<0.05$). The overall mortality rates of wild *An. gambiae s.l.* induced by organophosphate (fenitrothion and pirimiphos methyl) were significantly higher compared to other huts. More interestingly, pirimiphos methyl showed significantly the highest mortality rate within treatments (OR <1 and $p<0.001$).

At least 80% of blood fed *An. gambiae s.l.* died after taking rest on the treated wall in the hut treated with pirimiphos methyl. This will keep them from increasing their offspring. Furthermore, community-wide use of pirimiphos methyl in IRS in some circumstances will produce a 'mass effect' on the reduction of the density

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of infective mosquitoes in the area and, consequently, protect the whole community including those whose houses are not treated.

Among the insecticides tested in this study, those which performed well on walls were, respectively, pirimiphos methyl (4 to 5 months residual effect), lambdacyhalothrin (3 to 4 months) and bendiocarb (2 to 3 months residuel effect). All insecticides killed mosquitoes until the 4th month except pirimiphos methyl which still killed until the 6th month. Fenitrothion has showed the shortest, 1 to 3 months, residual effect. The duration of all of these insecticides is within the range of two to six months reported in WHO recommendations.[22] In fact, the spraying cycles may not exceed 13 weeks for bendiocarb on mud walls while it may last 13 weeks at least for the other types of surface, 20 weeks for lambdacyhalothrin on cement and mud walls and above 26 weeks for lambdacyhalothrin on wood walls.[23] ACTELLIC 50 EC (Syngenta UK Ltd, Fulbourn, Cambridge, UK) (pirimiphos methyl) used in evaluation in Ghana showed high levels of vector susceptibility for period of roughly 4 months, exceeding the effective minimum duration action of 2–3 months recommended by WHOPES.[24] A similar trial conducted in Mozambique demonstrated the effectiveness of ACTELLIC 50 EC for the control of *An. arabiensis* for at least 5 months on diverse surfaces. It is therefore recommended for use in IRS in Ghana and may be rotated with carbamate as a strategy for managing vector resistance.[24]

Finally, fenitrothion and pirimiphos methyl are the best potential candidate alternative insecticides to bendiocarb in IRS implementation to controlling Anopheles resistance to pyrethroid. Indeed, fenitrothion because of its persistent odour, has greatly reduced mosquitos' entry and has also induced tremendous feeding inhibition in mosquitoes. But pirimiphos methyl has induced a high mortality rate in mosquitoes combined with a long lasting residual effect on the walls. Based on key entomological parameters as deterrence and insecticidal activities, pirimiphos methyl can divert or kill mosquitoes before they can feed. Fenitrothion showed the best indicators, but its lethal effect cannot effectively exceed 2 months. However, at high community-level coverage, pirimiphos methyl can provide a high reduction in mosquito density and survival than a mere personal protection. Furthermore, its lethal effect can last for 4 months.

As a result, unlike, with organophosphates, there was significant survival (at least 20%) until after a blood meal was taken and this could be a drawback in controlling the older, infective mosquitoes. In spite of the decrease of the repellent and killer effects of pyrethroids on wild resistant *An. gambiae s.l.*, this compound can still divert, provide less individual protection and kill fewer resistant malaria vectors. However, their use in mosaic or in combination with other effective insecticide classes with different modes of action can have more impact in malaria vector control success. There was a lack of linkage between effectiveness and resistance; however, we underline that the study on 'Tracking and monitoring of insecticides resistance over sentinel sites' are carrying out.[24] It is also important to fill gaps in our knowledge base about the mechanisms of insecticide resistance, to study the impact of resistance on malaria control, and how our strategies for prevention are working.

Conclusion

After six months experiment of IRS treatments in experimental huts on *An. gambiae s.l.*, the organophosphates (pirimiphos methyl, fenitrothion WP 40%) and bendiocarb WP 80% appeared to be the most effective insecticides to controlling pyrethroid resistant Anopheles. Bendiocarb and fenitrothion decayed in less than four months, displaying a short lifespan on cement experimental walls. Nonetheless, bendiocarb still seems a promising insecticide to controlling resistant malaria vectors in Benin. However, to more success IRS in Atacora district, Benin NMCP has selected pirimiphos methyl as insecticide B to alternate or combine with bendiocarb (carbamate).

Authors' contributions: FRA and MCA designed the study. FRA, RA, RG, OO carried out the experiment. FRA, RA and FO analyzed the data. FRA drafted the manuscript. FRA and MCA critically revised the manuscript. All authors read and approved the manuscript. FRA is guarantor of the paper.

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